YUE, Wenchao

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Education and Training

- **The Chinese University of Hong Kong - CUHK** — Shatin, Hong Kong
  Doctor of Philosophy - Ph.D., Electronic Engineering — *August 2021 to November 2026 (expected)*
  Research Interest: *Medical Mechatronics Laboratory - Robotics, Perception and AI (RPAI) Group co-supervised by Prof. REN Hongliang (CUHK & NUS), Prof. YUAN Wu, Scott (CUHK).*

- **Swiss Federal Institute of Technology in Lausanne - EPFL** — Vaud, Lausanne
  Visiting Ph.D., Robotics, Control and Intelligent System — *November 2024 to July 2025*
  Research Interest: *Medical Origami Robotics For Haptic Rendering - Reconfigurable Robotics Lab (RRL) supervised by Prof. Jamie Paik.*

- **National University of Singapore - NUS** — Singapore
  M.S. Honors (Distinction), Mechanical Engineering - GPA: 4.5/5.0 — *August 2020 to June 2021*
  Master Thesis: *Quadruped Search and Rescue System Based on Hybrid Deep Reinforcement Learning Auxiliary Control Strategy. (Excellent Thesis: A) Hardware-based on Stanford Pupper- Supervised by Prof. TAY Eng Hock, Francis*

- **Institut Polytechnique des Sciences Avancées - IPSA** — Paris, France
  Exchange Student, Airframe and Materials - GPA: 19/20 (A) — *July 2019 to February 2020*
  The Only selected candidate among Mechanical Engineering students (academic year 2019/2020), fully funded by China Scholarship Council (CSC)

- **Nanjing University of Aeronautics and Astronautics - NUAA** — Nanjing, China
  B.E. Mechanical Engineering - GPA: 88/100 (GPA: 3.8) - Ranking: 2/34 — *August 2016 to June 2020*
  Bachelor Thesis: *Autonomous Palpation for Tumor Localization: Design of a Dynamic Tactile Sensor and Gaussian Process Adaptive Sampling (Excellent Thesis: A)*
  Activity: *Championship (the 1st ranking) in the 10th Undergraduate Academic Forum; Project Leader in Undergraduate Innovative Training Program; Core Member & co-author of the Engineering Practice project launched by Chengdu Aircraft Industrial (Group) & NUAA*

Research Key Words

Embodied AI, **Tactile Sensing**, **Robotic Dexterous Manipulation**, **Force/Tactile-VLA**, **Soft & Stretchable Electronics**, **Medical Mechatronics**

Research Experience

- **Sharpa Robotics** — Shanghai, China
  Foundation Model Research Intern — *July. 2026 - Present*
  - **Multi-Stage Tactile Modality Integration**: Led the tactile modality embedding and integration pipeline for embodied foundation models based on GR00T N1.7 architecture. Designed hierarchical cross-modal alignment paradigms spanning pre-training (FTP-1) and mid-training (T-Rex) stages, projecting high-dimensional tactile sensory signals into a unified vision-language latent space. Constructed robust cross-modal feature representation mechanisms to bridge tactile perception and visual-language semantics for embodied intelligent systems.
  - **Post-training Tactile Injection Fine-Tuning**: Proposed and engineered a novel second-order generalized tactile injection fine-tuning paradigm for model post-training, fully compatible with mainstream embodied frameworks including Pi05 and World Language Action (WLA) model. Optimized policy network architectures to dynamically activate and aggregate tactile attention features from foundation model backbones, empowering robots with precise force adjustment and fine-grained contact manipulation capabilities in complex physical interactive scenarios. (CVPR 2027 under submission)
  - **Dexterous Manipulation Benchmark & Generalization Evaluation**: Constructed a standardized multi-modal tactile-vision evaluation benchmark for dexterous robotic manipulation. Conducted systematic generalization validation of embodied foundation model policies across diverse contact-sensitive manipulation tasks. Significantly enhanced the model's zero-shot generalization ability and task success rate under unseen physical contact and environmental scenarios.

- **MindOn Robotics (Shenzhen)** — Shenzhen, China
  Embodied AI Algorithm Research Intern — *Apr. 2026 - Jul. 2026*
  - **VLA & Force-Tactile Multimodal Fusion**: Addressed core drawbacks of Vision-Language-Action (VLA) models in high-contact manipulation tasks, including tape sealing and GPU power cable insertion, such as visual blind zones, unstable contact force control, and low task success rate; implemented a fusion pipeline for vision-language models and force/tactile perception.
  - **Dual-Layer Force Compensation Control Design**: Fused tactile and force signals in both training and inference stages; established mapping relations among force sensing data, robot joint stiffness, and virtual trajectories; proposed a two-stage compensation control scheme with dynamic stiffness tuning and virtual target trajectory correction.

- **Full Pipeline Real Robot Deployment**: Built an end-to-end workflow covering data collection, model training, and real-time inference for vision-language-tactile fusion, and fully deployed the algorithm onto physical robots. Achieved on-demand autonomous force compensation during contact interactions, greatly boosting tape fitting accuracy and overall success rate of GPU cable plug-and-pull manipulation tasks.

- **The Chinese University of Hong Kong - Department of Electronic Engineering** Hong Kong
  Graduate Research Assistant & Teaching Assistant (Full-time) *August 2021 - Now*
  - **Graduate Research Assistant**: Guides master candidates in tactile-related final projects, which included the fabrication of flexible tactile sensors, the perception and acquisition of hardness distribution maps, etc.
  - **Teaching Assistant**: Involves in the tutorial course, homework arrangement, and exam paper production for undergraduate courses - Probability for Engineers (ENGG2760B) and Robotic Perception and Intelligence (ELEG3103)
  - **Laboratory Materials Manager**: Responsible for collecting equipment to be purchased for the lab, helping create a shopping list, and making the purchase happen
  - **Robotics, Perception and AI Group**: supervised by **Prof. REN Hongliang**

- **National University of Singapore - Department of Mechanical Engineering** Singapore
  Research Engineer (Full-time) *March 2021 - July 2021*
  - **Soft Finger Fabrication**: More focused on soft robotics and soft actuators, established the demo of the SDM Finger with Dragon Skin 30 and Smooth Sil 950/960
  - **NUS Soft Robotics Team**: supervised by **Prof. LASCHI Cecilia**

- **National University of Singapore - Department of Biomedical Engineering** Singapore
  Research Assistant (Part-time) *August 2020 - March 2021*
  - **Development of Piezoelectric Sensor**: Utilizing piezo 5A patches functioned as both actuator and sensor to propose a novel piezo sensor for the purpose of hardness measurement, which could be specifically used in the Minimally Invasive Surgery (MIS)/ Robot-assisted Minimally Invasive Surgery (RMIS).
  - **Undergraduate Mentor**: Guiding the undergraduate students at the National University of Singapore to operate actuating experiments based on the Piezo 5J patch to verify its feasibility (sensing & actuating).
  - **Medical Mechatronics Laboratory**: supervised by **Prof. REN Hongliang**

- **Nanjing University of Aeronautics and Astronautics - Mechatronics College** Nanjing, Jiangsu
  Research Assistant (Full-time) *May 2017 - August 2017*
  - **Literature Investigation**: Conducting and completing a literature review on Robot-assisted Minimally Invasive Surgical (RMIS) systems, tactile sensing for surgical robots, and tactile localization techniques in medical robotics systems
  - **Design and Simulation Study of Tactile Sensor**: Completing The structure design of the tactile sensor based on the piezoelectric effect independently. And further using ANSYS software to perform statics simulation, mode analysis, and harmonious response analysis.
  - **Fabrication and Experiment of Tactile Sensor**: Completing the fabrication, processing, and assembly of the sensor prototype, and building the experimental platform independently for testing and calibration. And, finally verifying that the proposed sensor has the function of hardness recognition by fresh pig liver.
  - **Mechatronics Laboratory**: supervised by **Prof. JU Feng**

- **Harbin Institute of Technology - Department of Computer Science** Harbin, Heilongjiang
  Research Assistant (Full-time) *May 2017 - August 2017*
  - **Humanoid Robot Design**: Finished the entire design of the humanoid robot with other three members, assembled the robot, and performed its gait adjustment and dance programming independently.
  - **Static and Dynamics Gait Study**: Mainly studied the gait planning algorithm, trajectory planning of robot motion, ZMP index, and the destabilizing factors of a biped robot, and successfully generated various static and dynamic gaits.
  - **ROS Programming**: Probed into the problem-solving, logical reasoning and definition deduction, obstacle avoidance, navigation, grabbing objects, object recognition, etc.
  - **Multi-intelligent Body Robotics Research Center**: supervised by **Prof. PIAO Songhao**

Professional Qualifications

- **Reviewers for Journals**: Cyborg and Bionic Systems, IEEE T-RO, IEEE/ASME T-MECH, IEEE T-II, IEEE RA-L, IEEE Sensors Journal, Advanced Robotics Research, Sensors, Biomimetic Intelligence and Robotics, ACS Applied Materials and Interfaces.
- **Reviewers for Conferences**: RoboSoft 26', ICRA 25', IROS 25', RoboSoft 25', ICIA 25', ICBIR 25', ROBIO 24', ICRA 24', IROS 24'.
- **Session Chair for Conferences**: ICBIR 25', ROBIO 24'.

Publications

- **Peer-Reviewed Conference/Journal Publications**:
  1. **W. Yue**†, F. Leng†, S.A. Abbasi†, Y. Hu, R. Tang, Y. Tian, X. Deng, W. Yuan*, C. Xu*, and H. Ren*, "MechDyna: Concurrent On-Needle Mechanoperception and Dynamic In-Situ Delivery Manipulation via Portable Thermoresponsive Piezoelectric Microneedle Patches", in **Nature Biomedical Engineering** (Under Submission)
  2. **W. Yue**, C. Xu, T. Zhang, J. Qiu, W. Yuan, H. Ren, "Bimodal Tactile Tomography with Bayesian Sequential Palpation" in **Cyborg and Bionic Systems. [IF: 18.2 - 2025]**.
  3. **W. Yue**, Z. Wang, Y. Wang, S. Demirtas, Y. Jiang, C. M. Lim, J. Paik, H. Ren, "Learning-Based Robotic Ultrasound Inference for Active Incision Observation Optimum in Pre-tracheostomy Planning" in **IEEE Transactions on Robotics** (Under 2nd Round Review)
  4. **W. Yue**, R. Zhang, K. Sun†, H. C. Cheung, R. Zhao, Y. Zhang, C. Xu, W. Yuan*, and H. Ren, "Bio-inspired Self-Oscillating Needle Transducer with Unified In-Keyhole Sensing and Actuation Capability" in **The 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems** (Under Review)
  5. **W. Yue**†, Y. Wang†, S. Chen, H. Gao, H. Ren, "Multimodal Augmented Reality Assisted Incision Guidance for Preoperative Tracheostomy Planning", in **The IEEE International Conference on Robotics and Biomimetics (ROBIO). [Best Student Paper Finalist]**.
  6. **W. Yue**, C. Bai, J. Lai, H. Ren*, "Dual-Stroke Soft Peltier Pouch Motor Based on Pipeless Thermo-Pneumatic Actuation" in **Advanced Engineering Materials**, 2301408. **[Cover Invited]**
  7. **W. Yue**, F. Bai, J. Liu, F. Ju, M. Q. H. Meng, C. M. Lim, H. Ren*, "RASEC: Rescaling Acquisition Strategy with Energy Constraints under SE-OU Fusion Kernel for Active Trachea Palpation and Incision Recommendation in Laryngeal Region", in **IEEE Transactions on Automation Science and Engineering**. **[ICRA 2025 Oral Invitation]**
  8. **W. Yue**, C. Xu, W. Yuan*, H. Ren*, "Miniature OCT-based Tactile Sensor for Concurrent Multimodal Contact Profiling in Robotic Intraluminal Palpation", in **ICRA 2025 Workshop - The Evolving Landscape of Surgical Robotics (ELSR)**. **[Best Paper Award and Best Poster Runner-Up Award]**
  9. **W. Yue**, C. Xu, W. Yuan*, H. Ren*, "Bimodal Tactile Tomography with Bayesian Sequential Palpation for Intracavitary Microstructure Profiling and Segmentation", in **ICRA 2025 Workshop - Robot-Assisted Medical Imaging (RAMI)**
  10. J. Lai, T. A. Ren, **W. Yue**, S. Su, J. Y. Chan, H. Ren*, (2023). "Sim-to-real transfer of soft robotic navigation strategies that learns from the virtual eye-in-hand vision", in **IEEE Transactions on Industrial Informatics**.
  11. **W. Yue**, S. Chen, K. Yan, Y. Wen, R. Tang, G. Hu*, and H. Ren*, "Soft Balloon Actuators Embodied Flexible Ink-Transfer-Printing Sensing for Self-Awareness Inflation and Microneedle Punch", in **Advanced Robotics Research**.
  12. **W. Yue**, P. Liu, J. Lai, C. Xu, W. Yuan*, and H. Ren*, "ElastoSight: Miniature Tactile Sensors for Multimodal Force and Geometry Estimation in Robotic Intraluminal Palpation", in **IEEE-RAS International Conference on Soft Robotics**.
  13. **W. Yue**, F. Ju, X. Zhang, Y. Yun, T. Li, Z. T. Tse, H. Ren*, (2023). "Dynamic Piezoelectric Tactile Sensing using Symmetrical Structure of Flexure Hinges and Anisotropic Vibration Modes", in **IEEE Sensors Journal**, vol. 21, no. 16, pp. 17712-17722, 15, Aug 2021, doi: 10.1109/JSEN.2021.3086114.
  14. S. Chen†, **W. Yue**†, R. Zhao, and H. Ren*, "Kirigami-inspired Flexible Strain Sensor for Dynamic Inflation Surface Applications", in **IEEE-RAS International Conference on Soft Robotics**.
  15. **W. Yue**, J. Qi, X. Song, S. Fan, H. Ren*, "Origami-Inspired Structure with Pneumatic-Induced Variable Stiffness for Multi-DOF Force-Sensing", in **Sensors**, vol. 22, no. 14, pp. 5370.
  16. L. Li†, **W. Yue**†, H. Ren*, " Surgical Force Haptic Feedback Technology", in **Handbook of Robotic Surgery**.
  17. Y. Zhang, X. Wei, **W. Yue**, C. Zhu, F. Ju*, "A dual-mode tactile hardness sensor for intraoperative tumor detection and tactile imaging in robot-assisted minimally invasive surgery", in **Smart Materials and Structures**, 16, July, 2021, 30(8), 085041.
  18. F. Ju*, H. Ye, D. Bai, Y. Zhang, C. Zhu, Y. Cao, **W. Yue**, "Autonomous tumor palpation and resection path planning using tactile array sensor and deep reinforcement learning for surgical robot", In **IEEE International Conference on Robotics and Biomimetics** (pp. 871-876).
  19. F. Ju*, Y. Hong, D. Bai, Z. Yin, C. Zhu, **W. Yue**, "A Laminated Continuum Robot with Gripping Force-Sensing Forceps for Robot-Assisted Minimally Invasive Surgeries", in **IEEE Sensors Journal**.
- **Patent Application**:
  1. **Yue wenchao**, Ju Feng, "A piezoelectric tactile sensor based on the symmetric structure of flexible hinge", accepted, Chinese Utility Model Patent application No.: 202004455H.
  2. **Yue wenchao**, Zhao Ruizhou, Ren Hongliang, "A Kirigami-inspired Flexible Strain Sensor for Dynamic Inflation Surface Applications", Chinese Utility Model Patent application, under submission

Proposal Writing Experience

- **Origami-Inspired Versatile Micromanipulator for Robot-assisted Surgery**:
  Major Contributor: Yue Wenchao. PIs: Prof. Ren Hongliang (CUHK) and Yue Wenchao
  Funding Details: IdeaBooster Fund [Equivalent to 25, 796 USD]
  Status: Submitted.

- **Human-Robot Interaction Co-integration of Flexible Bionic Inertial Perception Interface and Infection Prevention Assisted Medical Robot System**:
  Major Contributor: Yue Wenchao. PIs: Prof. Ren Hongliang (CUHK and CUHK-SZRI) and Prof. Hu Guohua (CUHK)
  Funding Details: Basic and Applied Basic Research Fund in Guangdong Province [Equivalent to 153, 846 USD]
  Status: Funded.

- **Robotic Endobronchial Multi-modal Biopsy of Pulmonary Nodules**:
  Subsection Author: Yue Wenchao. PIs: Prof. Ren Hongliang (CUHK), Prof. Xu Chenjie (CityU), and Prof. Chen Chia-Hung (CityU)
  Funding Details: Research Impact Fund (RIF) by University Grants Committee [Equivalent to 1, 163, 500 USD]

Status: Funded.

- **Minimally Contiguous Intubation and Tracheostomy for Severe COVID-19 Patients with Teleoperated Endotracheal cum Percutaneous Dual Robotic Modules and Multi-Modality Guidance**:
  Subsection Author: Yue Wenchao. PIs: Prof. Ren Hongliang (CUHK), Prof. Yuan Wu, Scott (CUHK), and Prof. Kwok Ka-Wai (HKU)
  Funding Details: Collaborative Research Fund (CRF) by University Grants Committee [Equivalent to 936, 000 USD]
  Status: Funded.

Academic Honors and Scholarship

- HKSAR Talent Development Award (10,000 HKD) - July. 2025
- Overseas Research Attachment Programme (ORAP) (approx. 50,000 HKD) - July. 2025
- PhD International Mobility for Partnerships and Collaborations Award (IMPAC Award) (approx. 50,000 HKD) - July. 2024
- The Second-place Award of The 8th Chinese Annual Seminar on Soft Robotic Theory and Technology - Nov. 2023
- Silver Award of the 13th Challenge Cup Chinese Student Entrepreneurship Project Competition 2023 (Hong Kong SAR Region Delegate Team) - March. 2023
- Fourth-place award - Project "AI-based Pulmonary Tree-branch Reconstruction for Precise Motion Planning in Personalized Robotic-assisted Bronchoscope Intervention" in AI x HK OpenCup 2022 - November. 2022
- Third-place Award - Project "Magnetic Capsule Robots" and Merit Award - Project "Modular Parallel Robot" in the 8th Hong Kong University Student Innovation and Entrepreneurship Competition - May. 2022
- Merit Award, the Engineering Medical Innovation Global Competition (EMedIC Global) - November. 2021
- Hong Kong Postgraduate Scholarship (PGS), Graduate School of The Chinese University of Hong Kong (approx. 886,000 HKD). - June. 2021
- Excellent Graduation Thesis, awarded by Innovative Final Year Project Fund in Jiangsu Province (5,000 RMB per semester, one-year funding) - June. 2020
- First prize in Advanced Mathematics Competition, Chinese Mathematical Society - July. 2020
- Excellent Graduation Thesis, Nanjing University of Aeronautics and Astronautics - June. 2020
- Champion of the 10th Undergraduate Academic Forum, Nanjing University of Aeronautics and Astronautics - December. 2019
- Second prize in Advanced Mathematics Competition, Chinese Mathematical Society. - July. 2019
- Excellent Undergraduate Exchange Scholarship, Institut Polytechnique des Sciences Avancees – IPSA (Paris), China Scholarship Council (1050 euro per month, seven-month funding) - June. 2019
- First prize in the Undergraduates' Engineering Integrative Ability Competition - March. 2019
- First prize of Spring Academic Scholarship, Nanjing University of Aeronautics and Astronautics - 2017 - 2020
- First prize of Excellent Scholarship, Nanjing University of Aeronautics and Astronautics - 2017 - 2020
- Second prize in the National Robot and Artificial Intelligence Competition, Humanoid Robot Group - Oct. 2017

Skills Summary

- **Languages**: Python, C, C++
- **Software**: Matlab, ANSYS, Abaqus, SolidWorks, AutoCAD, CATIA, COSMOL
- **Platforms**: Linux, Windows, Arduino, Raspberry